Online Flight Reservation System

AIM

To design and develop an Online Flight Reservation System using HTML, CSS, and JavaScript that allows users to search flights, book tickets, make payments, view booking status, cancel tickets, print tickets, and submit feedback. The system also provides an admin panel to manage flights, bookings, and reports.

CODE :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Flight Reservation System</title>

  <style>
    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #eef3f8;
      color: #222;
    }

    header {
      background: #063970;
      color: white;
      padding: 22px;
      text-align: center;
    }

    nav {
      background: #092c52;
```

```
  display: flex;
  justify-content: center;
  flex-wrap: wrap;
}

nav button {
  background: none;
  color: white;
  border: none;
  padding: 14px 16px;
  cursor: pointer;
}

nav button:hover {
  background: #1768ac;
}

.container {
  max-width: 900px;
  margin: 25px auto;
  padding: 20px;
}

.page {
  display: none;
  background: white;
  padding: 25px;
  border-radius: 10px;
  box-shadow: 0 3px 10px #ccc;
}

.page.active {
  display: block;
}
```

```
h2 {
  color: #063970;
}

input, select, textarea {
  width: 100%;
  padding: 11px;
  margin: 8px 0 15px;
  border: 1px solid #bbb;
  border-radius: 5px;
}

button.action {
  background: #0878c9;
  color: white;
  padding: 11px 18px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
  margin: 4px;
}

button.action:hover {
  background: #063970;
}

.flight {
  border: 1px solid #ddd;
  padding: 15px;
  margin: 12px 0;
  border-radius: 7px;
}
```

```
.flight h3 {
  color: #0878c9;
}

.message {
  color: #087443;
  font-weight: bold;
}

footer {
  text-align: center;
  background: #063970;
  color: white;
  padding: 15px;
  margin-top: 30px;
}

@media (max-width: 600px) {
  .container {
    margin: 10px;
    padding: 10px;
  }

  nav button {
    padding: 10px;
  }
}

@media print {
  header, nav, footer, .no-print {
    display: none;
  }

  .page {
```

```
        display: none !important;
        box-shadow: none;
      }

      #printTicket {
        display: block !important;
      }
    }
  </style>
</head>

<body>

<header>
  <h1>Online Flight Reservation System</h1>
  <p>Search Flights | Book Tickets | Manage Reservations</p>
</header>

<nav>
  <button onclick="showPage('search')">Search Flight</button>
  <button onclick="showPage('login')">Admin Login</button>
  <button onclick="showPage('forgot')">Forgot Password</button>
  <button onclick="showPage('booking')">Book Ticket</button>
  <button onclick="showPage('payment')">Payment</button>
  <button onclick="showPage('myticket')">My Ticket</button>
  <button onclick="showPage('cancel')">Cancel Ticket</button>
  <button onclick="showPage('feedback')">Feedback</button>
  <button onclick="showPage('report')">Admin Report</button>
</nav>

<div class="container">

  <!-- Search Flight Page -->
  <section id="search" class="page active">
```

```
  <h2>Search Flight</h2>

  <label>From</label>
  <input type="text" id="from" placeholder="Enter departure city">

  <label>Destination</label>
  <input type="text" id="to" placeholder="Enter destination city">

  <label>Travel Date</label>
  <input type="date" id="date">

  <label>Travel Class</label>
  <select id="class">
    <option>Economy</option>
    <option>Business</option>
    <option>First Class</option>
  </select>

  <button class="action" onclick="searchFlights()">Search Flights</button>

  <div id="flightResults"></div>
</section>

<!-- Administrator Login Page -->
<section id="login" class="page">
  <h2>Administrator Login</h2>

  <label>Username</label>
  <input type="text" id="adminUser" placeholder="Enter username">

  <label>Password</label>
  <input type="password" id="adminPass" placeholder="Enter password">

  <button class="action" onclick="adminLogin()">Login</button>
```

```
  <p id="loginMessage" class="message"></p>
  <p>Demo username: admin</p>
  <p>Demo password: admin123</p>
</section>

<!-- Forgot Password Page -->
<section id="forgot" class="page">
  <h2>Forgot Password</h2>

  <label>Registered Email</label>
  <input type="email" id="forgotEmail" placeholder="Enter your email">

  <button class="action" onclick="forgotPassword()">Reset Password</button>
  <p id="forgotMessage" class="message"></p>
</section>

<!-- Book Ticket Page -->
<section id="booking" class="page">
  <h2>Book Flight Ticket</h2>

  <label>Selected Flight</label>
  <select id="selectedFlight">
    <option value="AI101">AI101 - Chennai to Delhi</option>
    <option value="AI202">AI202 - Delhi to Mumbai</option>
    <option value="AI303">AI303 - Chennai to Bengaluru</option>
  </select>

  <label>Passenger Name</label>
  <input type="text" id="passenger" placeholder="Enter passenger name">

  <label>Age</label>
  <input type="number" id="age" min="1" placeholder="Enter age">

  <label>Passenger Email</label>
```

```
    <input type="email" id="email" placeholder="Enter email">

    <label>Seat Number</label>
    <select id="seat">
      <option>1A</option>
      <option>1B</option>
      <option>2A</option>
      <option>2B</option>
      <option>3A</option>
      <option>3B</option>
    </select>

    <button class="action" onclick="bookTicket()">Confirm Booking</button>
    <p id="bookingMessage" class="message"></p>
  </section>

  <!-- Ticket Payment Page -->
  <section id="payment" class="page">
    <h2>Ticket Payment</h2>

    <label>PNR Number</label>
    <input type="text" id="paymentPNR" placeholder="Enter PNR">

    <label>Payment Method</label>
    <select id="paymentMethod">
      <option>Credit Card</option>
      <option>Debit Card</option>
      <option>UPI</option>
      <option>Net Banking</option>
    </select>

    <button class="action" onclick="makePayment()">Pay Now</button>
    <p id="paymentMessage" class="message"></p>
  </section>
```

```
<!-- My Ticket / Booking Status Page -->
<section id="myticket" class="page">
  <h2>My Ticket / Booking Status</h2>

  <label>Enter PNR Number</label>
  <input type="text" id="searchPNR" placeholder="Enter PNR number">

  <button class="action" onclick="viewTicket()">View Ticket</button>

  <div id="ticketDetails"></div>
</section>

<!-- Cancel Ticket Page -->
<section id="cancel" class="page">
  <h2>Cancel Ticket</h2>

  <label>PNR Number</label>
  <input type="text" id="cancelPNR" placeholder="Enter PNR number">

  <button class="action" onclick="cancelTicket()">Cancel Ticket</button>
  <p id="cancelMessage" class="message"></p>
</section>

<!-- Customer Feedback Page -->
<section id="feedback" class="page">
  <h2>Customer Feedback</h2>

  <label>Name</label>
  <input type="text" id="feedbackName" placeholder="Enter your name">

  <label>Rating</label>
  <select id="rating">
    <option>Excellent</option>
```

```
      <option>Good</option>
      <option>Average</option>
      <option>Poor</option>
    </select>

    <label>Comments</label>
    <textarea id="comments" rows="4"
          placeholder="Write your feedback"></textarea>

    <button class="action" onclick="submitFeedback()">Submit Feedback</button>
    <p id="feedbackMessage" class="message"></p>
  </section>

  <!-- Admin Report Page -->
  <section id="report" class="page">
    <h2>Admin Report</h2>

    <p>Total Bookings: <strong id="totalBookings">0</strong></p>
    <p>Confirmed Tickets: <strong id="confirmedBookings">0</strong></p>
    <p>Cancelled Tickets: <strong id="cancelledBookings">0</strong></p>

    <button class="action" onclick="showReport()">Refresh Report</button>
    <button class="action" onclick="showPage('manage')">Manage Flights</button>
  </section>

  <!-- Admin Flight Management -->
  <section id="manage" class="page">
    <h2>Manage Flight Schedules</h2>

    <label>Flight Number</label>
    <input type="text" id="newFlightNumber" placeholder="e.g. AI404">

    <label>Route</label>
    <input type="text" id="newRoute" placeholder="e.g. Hyderabad to Goa">
```

```
    <label>Fare (INR)</label>
    <input type="number" id="newFare" min="1" placeholder="Enter fare">

    <button class="action" onclick="addFlight()">Add Flight</button>
    <div id="manageResults"></div>
  </section>

  <!-- Print Ticket Page -->
  <section id="printTicket" class="page">
    <h2>Print E-Ticket</h2>
    <div id="printDetails">Search for a PNR to display ticket details.</div>
    <button class="action no-print" onclick="window.print()">Print Ticket</button>
  </section>

</div>

<footer>
  <p>Online Flight Reservation System | Internet Programming Lab</p>
</footer>

<script>
  let bookings = [];
  let flights = [
    { number: "AI101", route: "Chennai to Delhi", fare: 5500, seats: 30 },
    { number: "AI202", route: "Delhi to Mumbai", fare: 4200, seats: 25 },
    { number: "AI303", route: "Chennai to Bengaluru", fare: 2800, seats: 20 }
  ];

  let selectedPNR = "";
  let adminAuthenticated = false;

  function showPage(pageId) {
    document.querySelectorAll(".page").forEach(page => {
```

```
    page.classList.remove("active");
  });

  document.getElementById(pageId).classList.add("active");
}

function searchFlights() {
  const from = document.getElementById("from").value.trim();
  const to = document.getElementById("to").value.trim();
  const date = document.getElementById("date").value;
  const travelClass = document.getElementById("class").value;
  const results = document.getElementById("flightResults");

  if (!from || !to || !date) {
    alert("Please enter departure, destination, and travel date.");
    return;
  }

  const matchingFlights = flights.filter(flight =>
    flight.route.toLowerCase().includes(from.toLowerCase()) &&
    flight.route.toLowerCase().includes(to.toLowerCase())
  );

  if (matchingFlights.length === 0) {
    results.innerHTML = "<p>No matching demo flights found. Try Chennai to Delhi, Delhi to Mumbai, or Chennai to Bengaluru.</p>";
    return;
  }

  results.innerHTML = "<h3>Available Flights</h3>";

  matchingFlights.forEach(flight => {
    results.innerHTML += `
      <div class="flight">
```

```
            <h3>${flight.number}</h3>
            <p>Route: ${flight.route}</p>
            <p>Date: ${date}</p>
            <p>Class: ${travelClass}</p>
            <p>Demo seats available: ${flight.seats}</p>
            <p>Fare: ₹${flight.fare}</p>
            <button class="action"
              onclick="selectFlight('${flight.number}')">
              Book This Flight
            </button>
          </div>
        `;
    });
}

function selectFlight(flightNumber) {
    document.getElementById("selectedFlight").value = flightNumber;
    showPage("booking");
}

function bookTicket() {
    const flightNumber = document.getElementById("selectedFlight").value;
    const passenger = document.getElementById("passenger").value.trim();
    const age = document.getElementById("age").value;
    const email = document.getElementById("email").value.trim();
    const seat = document.getElementById("seat").value;

    const flight = flights.find(item => item.number === flightNumber);

    if (!passenger || !age || Number(age) < 1 || !email) {
        alert("Please enter valid passenger details.");
        return;
    }
```

```
if (!flight || flight.seats <= 0) {
  alert("No seats available on this flight.");
  return;
}

const pnr = "FL" + Math.floor(100000 + Math.random() * 900000);

const booking = {
  pnr,
  flightNumber,
  route: flight.route,
  passenger,
  age,
  email,
  seat,
  fare: flight.fare,
  status: "Confirmed",
  paid: false,
  date: document.getElementById("date").value || "Not specified"
};

bookings.push(booking);
flight.seats--;

selectedPNR = pnr;
document.getElementById("paymentPNR").value = pnr;
document.getElementById("searchPNR").value = pnr;
document.getElementById("cancelPNR").value = pnr;

document.getElementById("bookingMessage").innerHTML =
  "Booking created! Your PNR is " + pnr +
  ". Continue to payment to complete the demo booking.";

showReport();
```

```
}

function makePayment() {
  const pnr = document.getElementById("paymentPNR").value.trim();
  const booking = bookings.find(item => item.pnr === pnr);

  if (!booking || booking.status === "Cancelled") {
    document.getElementById("paymentMessage").textContent =
      "Valid active booking not found.";
    return;
  }

  if (booking.paid) {
    document.getElementById("paymentMessage").textContent =
      "Payment has already been recorded for this demo booking.";
    return;
  }

  booking.paid = true;

  document.getElementById("paymentMessage").textContent =
    "Demo payment recorded for ₹" + booking.fare +
    ". No real money has been charged.";

  showReport();
}

function viewTicket() {
  const pnr = document.getElementById("searchPNR").value.trim();
  const booking = bookings.find(item => item.pnr === pnr);
  const details = document.getElementById("ticketDetails");

  if (!booking) {
    details.innerHTML = "<p>Ticket not found. Please check the PNR.</p>";
```

```
    return;
  }

  const ticketHTML = `
    <div class="flight">
      <h3>Flight E-Ticket</h3>
      <p><strong>PNR:</strong> ${booking.pnr}</p>
      <p><strong>Passenger:</strong> ${booking.passenger}</p>
      <p><strong>Flight:</strong> ${booking.flightNumber}</p>
      <p><strong>Route:</strong> ${booking.route}</p>
      <p><strong>Travel Date:</strong> ${booking.date}</p>
      <p><strong>Seat:</strong> ${booking.seat}</p>
      <p><strong>Fare:</strong> ₹${booking.fare}</p>
      <p><strong>Payment:</strong> ${booking.paid ? "Demo payment recorded" : "Pending"}</p>
      <p><strong>Status:</strong> ${booking.status}</p>
      <button class="action"
        onclick="preparePrint('${booking.pnr}')">
        Print Ticket
      </button>
    </div>
  `;

  details.innerHTML = ticketHTML;
}

function preparePrint(pnr) {
  const booking = bookings.find(item => item.pnr === pnr);

  if (!booking) return;

  document.getElementById("printDetails").innerHTML = `
    <h3>ONLINE FLIGHT RESERVATION - E-TICKET</h3>
    <p>PNR: ${booking.pnr}</p>
    <p>Passenger: ${booking.passenger}</p>
```

```
    <p>Flight Number: ${booking.flightNumber}</p>
    <p>Route: ${booking.route}</p>
    <p>Travel Date: ${booking.date}</p>
    <p>Seat: ${booking.seat}</p>
    <p>Fare: ₹${booking.fare}</p>
    <p>Payment: ${booking.paid ? "Demo payment recorded" : "Pending"}</p>
    <p>Status: ${booking.status}</p>
    <p><em>This is a demonstration ticket, not a valid airline ticket.</em></p>
  `;

  showPage("printTicket");
  window.print();
}

function cancelTicket() {
  const pnr = document.getElementById("cancelPNR").value.trim();
  const booking = bookings.find(item => item.pnr === pnr);

  if (!booking) {
    document.getElementById("cancelMessage").textContent =
      "Ticket not found.";
    return;
  }

  if (booking.status === "Cancelled") {
    document.getElementById("cancelMessage").textContent =
      "This ticket is already cancelled.";
    return;
  }

  booking.status = "Cancelled";

  const flight = flights.find(item => item.number === booking.flightNumber);
  if (flight) flight.seats++;
```

```
  document.getElementById("cancelMessage").textContent =
    "Demo ticket cancelled. No real refund has been processed.";

  showReport();
}

function submitFeedback() {
  const name = document.getElementById("feedbackName").value.trim();
  const comments = document.getElementById("comments").value.trim();

  if (!name || !comments) {
    alert("Please enter your name and feedback.");
    return;
  }

  document.getElementById("feedbackMessage").textContent =
    "Thank you, " + name + "! Your demo feedback has been submitted.";

  document.getElementById("feedbackName").value = "";
  document.getElementById("comments").value = "";
}

function adminLogin() {
  const username = document.getElementById("adminUser").value;
  const password = document.getElementById("adminPass").value;

  if (username === "admin" && password === "admin123") {
    adminAuthenticated = true;
    document.getElementById("loginMessage").textContent =
      "Demo admin login successful.";
    showPage("report");
    showReport();
  } else {
```

```
    document.getElementById("loginMessage").textContent =
      "Invalid username or password.";
  }
}

function forgotPassword() {
  const email = document.getElementById("forgotEmail").value.trim();

  if (!email || !email.includes("@")) {
    document.getElementById("forgotMessage").textContent =
      "Please enter a valid email address.";
    return;
  }

  document.getElementById("forgotMessage").textContent =
    "Demo only: password reset email service is not connected.";
}

function showReport() {
  document.getElementById("totalBookings").textContent = bookings.length;

  document.getElementById("confirmedBookings").textContent =
    bookings.filter(item => item.status === "Confirmed").length;

  document.getElementById("cancelledBookings").textContent =
    bookings.filter(item => item.status === "Cancelled").length;
}

function addFlight() {
  if (!adminAuthenticated) {
    alert("Please log in as admin first.");
    showPage("login");
    return;
  }
```

```
  const number = document.getElementById("newFlightNumber").value.trim();
  const route = document.getElementById("newRoute").value.trim();
  const fare = Number(document.getElementById("newFare").value);

  if (!number || !route || fare <= 0) {
    alert("Enter a flight number, route, and valid fare.");
    return;
  }

  if (flights.some(item => item.number === number)) {
    alert("This flight number already exists.");
    return;
  }

  flights.push({ number, route, fare, seats: 30 });

  document.getElementById("manageResults").innerHTML =
    "<p class='message'>Flight added successfully.</p>";

  document.getElementById("newFlightNumber").value = "";
  document.getElementById("newRoute").value = "";
  document.getElementById("newFare").value = "";

  updateFlightOptions();
}

function updateFlightOptions() {
  const select = document.getElementById("selectedFlight");

  select.innerHTML = flights.map(flight =>
    `<option value="${flight.number}">${flight.number} - ${flight.route}</option>`
  ).join("");
}
```

```
  showReport();
</script>


</body>
</html>
```

